

Input Filter Design

Based on TI APP Note:SNVA538

<http://www.ti.com/lit/an/snva538/snva538.pdf>

All graphs are created with ZedGraph Plugin

+ Input Parameters

Undamped LC Filter

$$F_{cross} := 32 \text{ kHz}$$

$$f_{cutoff} := 5 \text{ kHz}$$

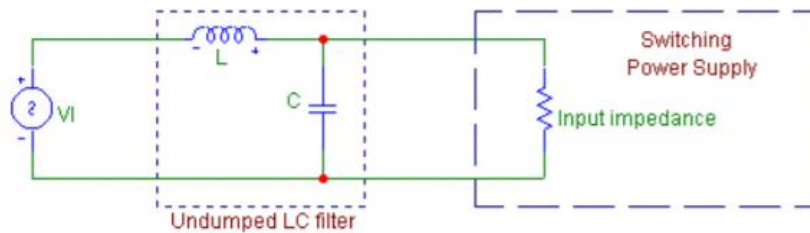
$$\omega C := f_{cutoff} \cdot 2 \cdot \pi = 31.4159 \text{ kHz}$$

$$R_{in} := 25 \text{ ohm}$$

Input cap of the Power Supply

$$C_{in_Supply} := 15 \text{ }\mu\text{F}$$

UNDAMPED LC FILTER



Undamped LC_Filter

$$L_{filter} := \frac{1}{\omega C^2 \cdot C_{in_Supply}} = 67.5475 \text{ }\mu\text{H}$$

$$\zeta := \frac{L_{filter}}{2 \cdot R_{in} \cdot \sqrt{L_{filter} \cdot C_{in_Supply}}} = 0.0424$$

selected L and C

$$L_{filter} := 33 \text{ }\mu\text{H}$$

$$C_{in} := 47 \text{ }\mu\text{F}$$

$$R_{dcr_Filter} := 30 \text{ m}\Omega$$

$$R_{ESR_C_in} := 150 \text{ m}\Omega$$

$$f_{C_filter1} := \frac{1}{2 \cdot \pi \cdot \sqrt{L_{filter} \cdot C_{in}}} = 4.0412 \text{ kHz}$$

Transfer Function

$$Z1_j := R_{dcr_Filter} + s_j \cdot L_{filter}$$

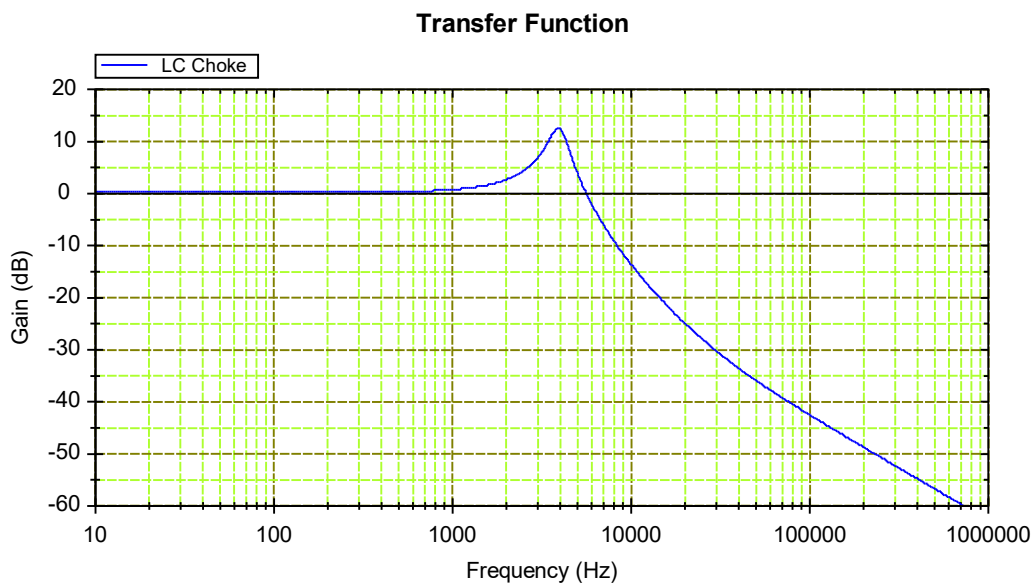
$$Z2_j := R_{ESR_C_in} + \frac{1}{s_j \cdot C_{in}}$$

$$Z2EQ_j := \frac{Z2_j \cdot R_{in}}{Z2_j + R_{in}}$$

$$LC_{filter}_j := \text{eval} \left((20) \cdot \log_{10} \left(\left| \frac{Z2EQ_j}{Z2EQ_j + Z1_j} \right| \right) \right)$$

$$\text{plot}_{transfer_f_LC} := \text{eval} \left(\text{augment} \left(\vec{f}, \overrightarrow{LC_{filter}} \right) \right)$$

Graph transfer function of LC filter

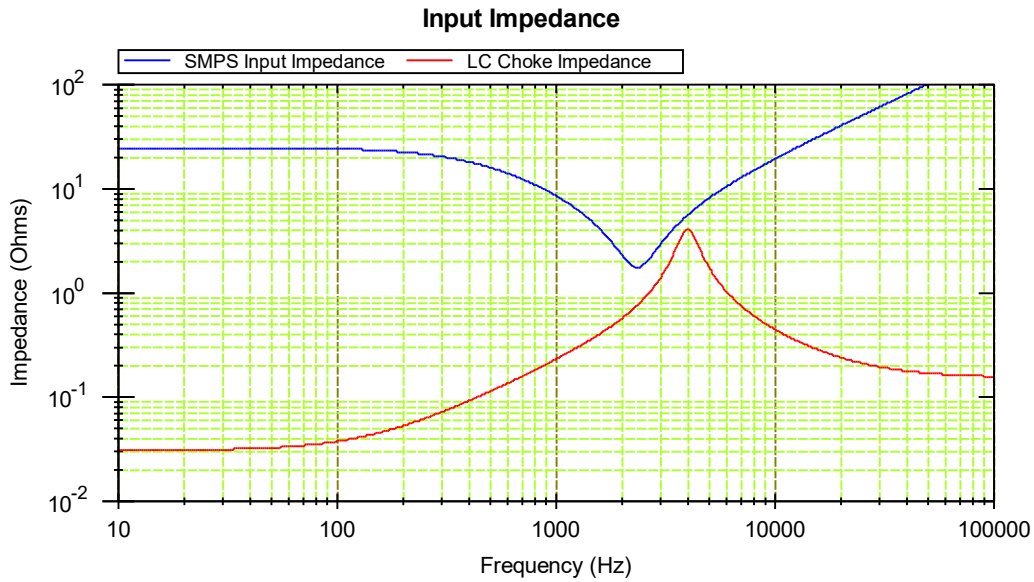


$$\text{plot}_{transfer_f_LC}$$

$$zf_j := \text{eval} \left(\left| \frac{Z1_j \cdot Z2_j}{Z1_j + Z2_j} \right| \right)$$

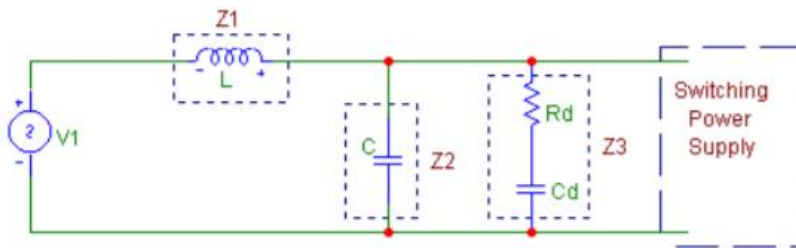
```
plotLC := eval (augment (f, Zf))
```

Graph input impedance vs Filter Output impedance



```
{plotinput_impedance
plotLC}
```

Parallel Damping



Parallel Damping

$$R_{parallel_damp} := \sqrt{\frac{L_{filter}}{C_{in}}} = 837.9306 \text{ m}\Omega$$

$$C_{damp} := 4 \cdot C_{in} = 188 \text{ }\mu\text{F}$$

$$R_{ESD_C_Damp} := 200 \text{ m}\Omega$$

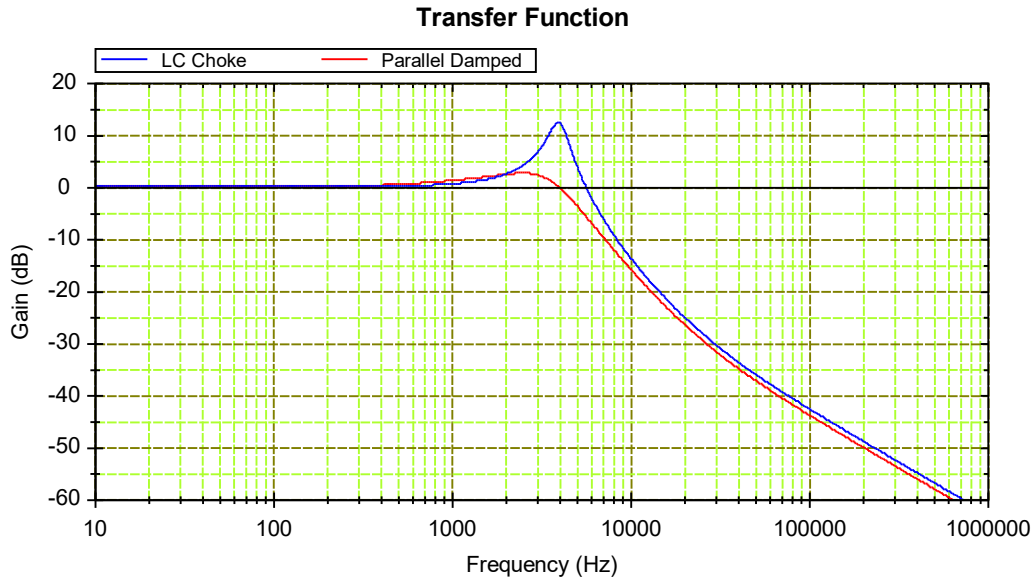
$$Z3_j := \frac{1}{s_j \cdot C_{damp}} + R_{ESD_C_Damp} + R_{parallel_damp}$$

$$Z3EQ_j := \frac{Z2EQ_j \cdot Z3_j}{Z2EQ_j + Z3_j}$$

Transfer Function of Parallel Damped LC Filter

$$Parallel_Damped_j := 20 \cdot \log_{10} \left(\left| \frac{Z3EQ_j}{Z3EQ_j + Z1_j} \right| \right)$$

```
plot_transfer_f_LC_PDAMP := eval (augment (f, Parallel_Damped))
```

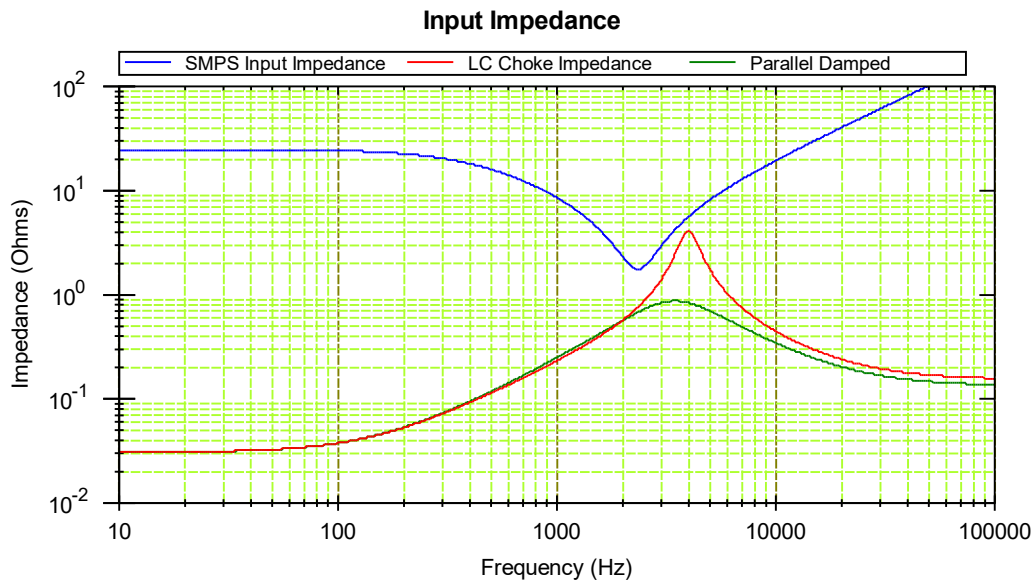


```
{ plot_transfer_f_LC
  plot_transfer_f_LC_PDAMP }
```

$$Zf2_j := \text{eval} \left(\left| \frac{Z1_j \cdot Z3EQ_j}{Z1_j + Z3EQ_j} \right| \right)$$

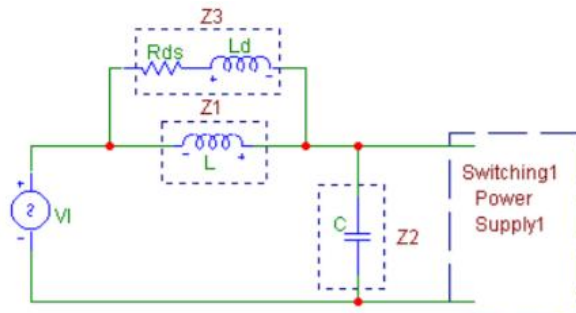
```
plot_P_damp := eval (augment (f, Zf2))
```

Graph input impedance vs Filter Output impedance



$\left\{ \begin{array}{l} \text{plot}_{input_impedance} \\ \text{plot}_{LC} \\ \text{plot}_{P_damp} \end{array} \right.$

Series Damped



Series Damping

$$n_{note_sd} := \frac{2}{15}$$

$$L_{series_damp} := L_{filter} \cdot n_{note_sd} = 4.4 \mu H$$

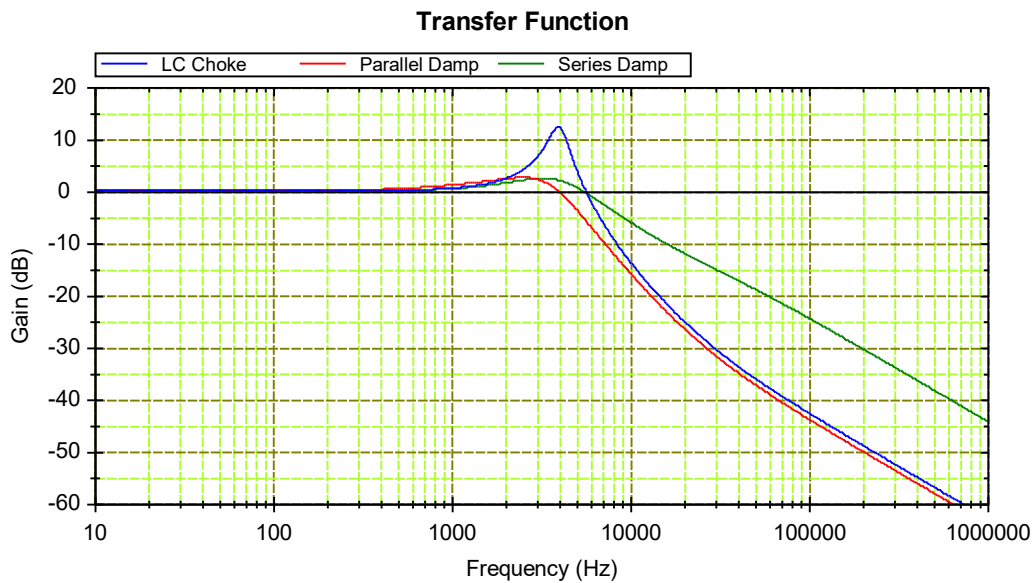
$$R_{SD_damp} := \frac{\sqrt{L_{filter}}}{\sqrt{C_{in}}} = 837.9306 m\Omega$$

$$Z3s_j := R_{SD_damp} + s_j \cdot L_{series_damp}$$

$$Z13_j := \frac{Z1_j \cdot Z3s_j}{Z1_j + Z3s_j}$$

$$Series_Damped_j := 20 \cdot \log_{10} \left(\left| \frac{Z2_j}{Z2_j + Z13_j} \right| \right)$$

$$plot_{transfer_f_LC_S_DAMP} := eval \left(augment \left(\vec{f}, \overrightarrow{Series_Damped} \right) \right)$$

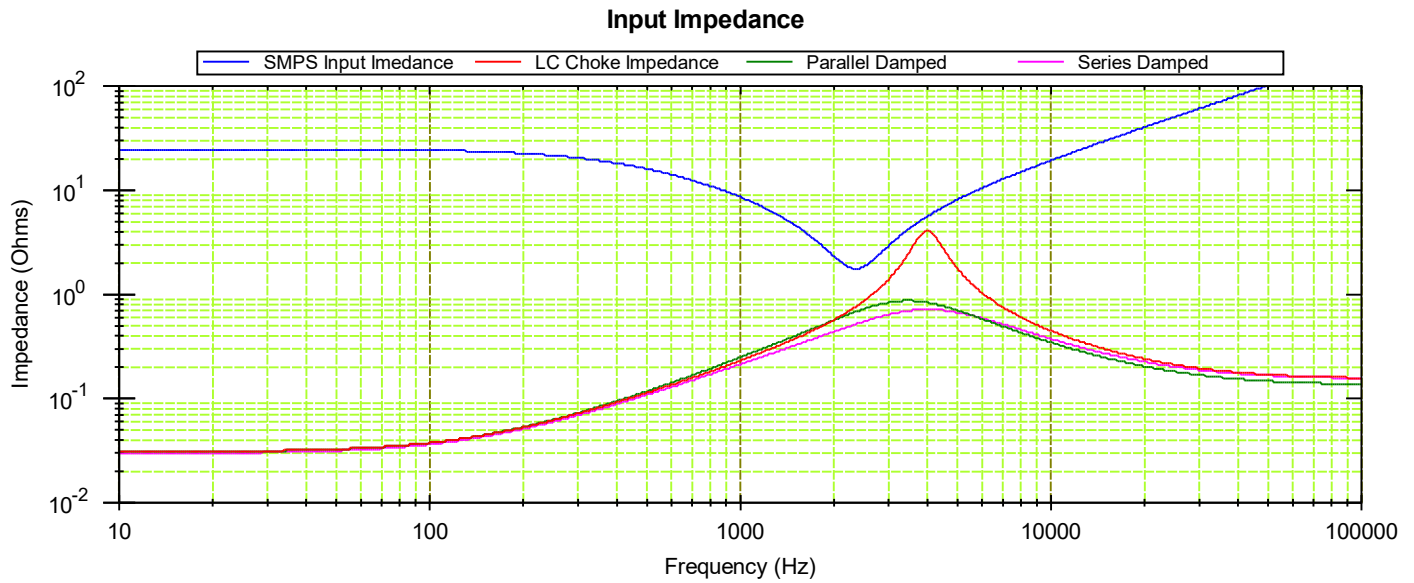


$$\begin{cases} plot_{transfer_f_LC} \\ plot_{transfer_f_LC_PDAMP} \\ plot_{transfer_f_LC_S_DAMP} \end{cases}$$

$$zf3_j := eval \left(\left| \frac{Z2_j \cdot Z13_j}{Z2_j + Z13_j} \right| \right)$$

$$plot_{s_damp} := eval \left(augment \left(\vec{f}, \overrightarrow{zf3} \right) \right)$$

Graph input impedance vs Filter Output impedance

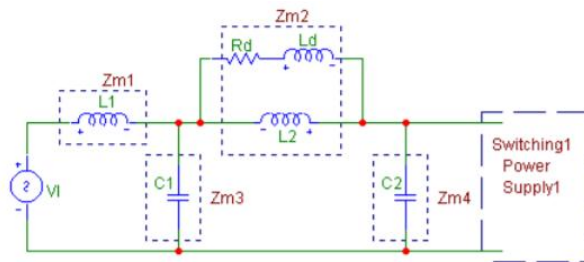


```

{
  plot input_impedance
  plot LC
  plot p_damp
  plot s_damp
}

```

Multiple Filter Sections



First Stage LC Filter

$$L1_{2stage} := \frac{L_{filter}}{4} = 8.25 \mu H$$

$$R_{dcr_L1_2stage} := 100 \text{ m}\Omega$$

$$C1_{2stage} := \frac{C_{in}}{4} = 11.75 \mu F$$

$$R_{C1_2stage_ESR} := 120 \text{ m}\Omega$$

$$f_{stage1_cutoff} := \frac{1}{2 \cdot \pi \cdot \sqrt{L1_{2stage} \cdot C1_{2stage}}} = 16.1649 \text{ kHz}$$

Second Stage LC Filter

$$L2_{2stage} := 7 \cdot L1_{2stage} = 57.75 \text{ } \mu\text{H}$$

$$R_{dcr_L2_2stage} := 100 \text{ m}\Omega$$

$$C2_{2stage} := 4 \cdot C1_{2stage} = 47 \text{ } \mu\text{F}$$

$$R_{C2_2stage_ESR} := 120 \text{ m}\Omega$$

$$f_{stage2_cutoff} := \frac{1}{2 \cdot \pi \cdot \sqrt{L2_{2stage} \cdot C2_{2stage}}} = 3.0549 \text{ kHz}$$

$$R_{damp_2stage} := \sqrt{\frac{L1_{2stage}}{C2_{2stage}}} = 418.9653 \text{ m}\Omega$$

$$Ld_{2stage} := \frac{L1_{2stage}}{8} = 1.0312 \text{ } \mu\text{H}$$

$$Zm1_j := s_j \cdot L1_{2stage} + R_{dcr_L1_2stage}$$

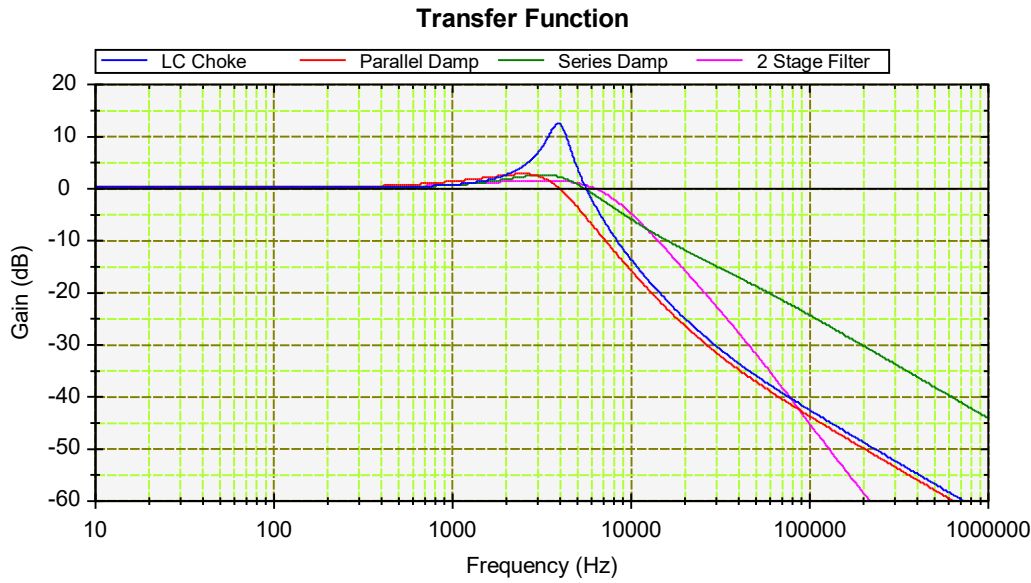
$$Zm2_j := \frac{1}{s_j \cdot C1_{2stage}} + R_{C1_2stage_ESR}$$

$$Zm3_j := \frac{(R_{damp_2stage} + s_j \cdot Ld_{2stage}) \cdot (s_j \cdot L2_{2stage} + R_{dcr_L2_2stage})}{(R_{damp_2stage} + s_j \cdot Ld_{2stage}) + s_j \cdot L2_{2stage} + R_{dcr_L2_2stage}}$$

$$Zm4_j := \frac{1}{s_j \cdot C2_{2stage}} + R_{C2_2stage_ESR}$$

$$\text{MultipleFilter}_j := \text{eval} \left(20 \cdot \log_{10} \left(\left| \frac{Z_{m4}_j}{\frac{Z_{m1}_j \cdot Z_{m2}_j}{Z_{m1}_j + Z_{m2}_j} + Z_{m3}_j + Z_{m4}_j} \cdot \frac{Z_{m2}_j}{Z_{m1}_j + Z_{m2}_j}} \right| \right) \right)$$

$\text{plot}_{\text{multiple_filter}} := \text{eval} \left(\text{augment} \left(\vec{f}, \overrightarrow{\text{MultipleFilter}} \right) \right)$

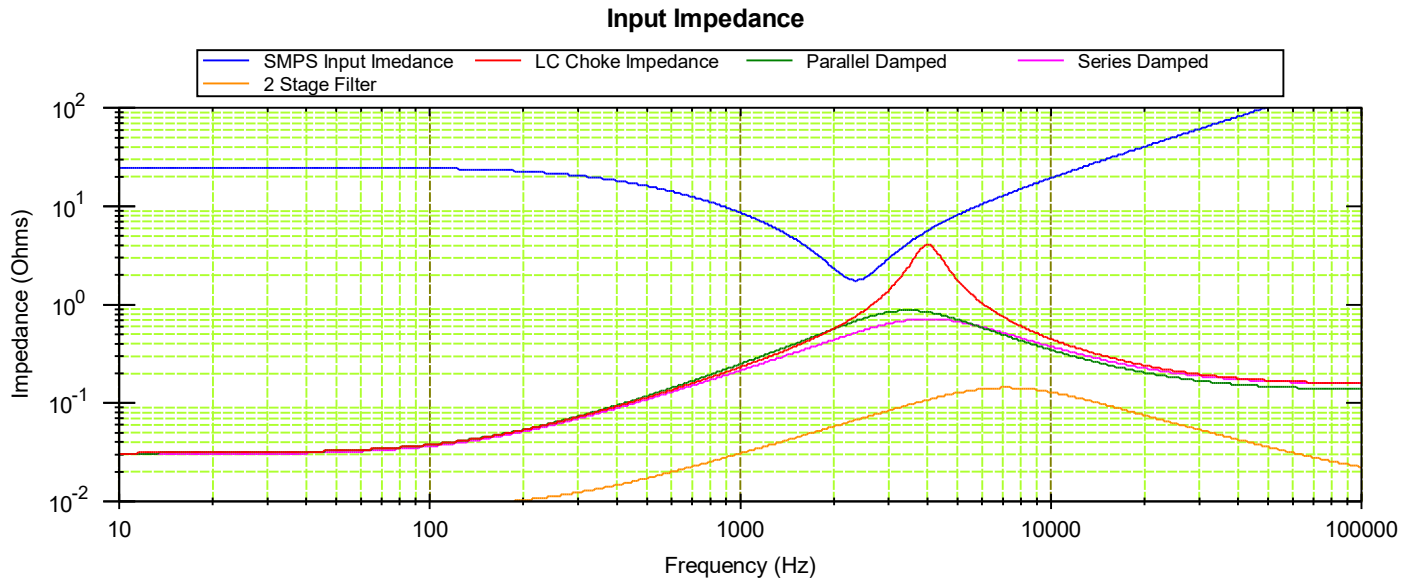


$\left\{ \begin{array}{l} \text{plot}_{\text{transfer_f_LC}} \\ \text{plot}_{\text{transfer_f_LC_PDAMP}} \\ \text{plot}_{\text{transfer_f_LC_S_DAMP}} \\ \text{plot}_{\text{multiple_filter}} \end{array} \right.$

$$Z_{f4}_j := \text{eval} \left(\left| \frac{Z_{m4}_j \cdot \frac{Z_{m1}_j \cdot Z_{m2}_j}{Z_{m1}_j + Z_{m2}_j} \cdot Z_{m3}_j}{\frac{Z_{m1}_j \cdot Z_{m2}_j}{Z_{m1}_j + Z_{m2}_j} + Z_{m3}_j + Z_{m4}_j} \right| \right)$$

$\text{plot}_{\text{multiple_filter_imp}} := \text{eval} \left(\text{augment} \left(\vec{f}, \overrightarrow{Z_{f4}} \right) \right)$

Graph input impedance vs Filter Output impedance



```
{ plot_input_impedance  
plot_LC  
plot_P_damp  
plot_S_damp  
plot_multiple_filter_imp
```